

Linear Functions & Inequalities

ANSWER KEY



Slope and slope-intercept form

- Find the slope of the line passing through (2, 3) and (6, 11).
$$\text{Slope} = \frac{11 - 3}{6 - 2} = \frac{8}{4} = 2.$$
- Write the equation of the line with slope -3 passing through (1, 4), in slope-intercept form.
 $y - 4 = -3(x - 1)$, so $y = -3x + 3 + 4 = -3x + 7$.

Parallel and perpendicular lines

- Find the equation of the line parallel to $y = 2x - 5$ passing through (3, 4).
Parallel lines share the same slope, $m = 2$. $y - 4 = 2(x - 3)$, so $y = 2x - 6 + 4 = 2x - 2$.
- Find the equation of the line perpendicular to $y = \frac{1}{2}x + 3$ passing through (4, 1).
The perpendicular slope is the negative reciprocal: $m = -2$. $y - 1 = -2(x - 4)$, so $y = -2x + 8 + 1 = -2x + 9$.

Finding equations from two points or a table

- A linear function passes through (0, 5) and (4, 17). Find its equation.
$$\text{Slope} = \frac{17 - 5}{4 - 0} = \frac{12}{4} = 3.$$
 Since (0, 5) is the y -intercept, the equation is $y = 3x + 5$.
- A table shows $x = 1, y = 7$ and $x = 3, y = 15$ for a linear function. Find the equation.
$$\text{Slope} = \frac{15 - 7}{3 - 1} = \frac{8}{2} = 4.$$
 Using point (1, 7): $y - 7 = 4(x - 1)$, so $y = 4x - 4 + 7 = 4x + 3$.

Solving and graphing linear inequalities

- Solve the inequality $3x - 4 < 11$, and graph the solution on a number line description.
 $3x < 15$, so $x < 5$. On a number line, this is an open circle at 5 with shading extending to the left (all values less than 5).
- Solve the inequality $-2x + 7 \geq 1$, being careful with the direction of the inequality sign.
 $-2x \geq -6$. Dividing by -2 flips the inequality: $x \leq 3$.

Systems of linear inequalities

- 9 A system requires $y > 2x - 1$ and $y \leq -x + 5$. Determine whether the point $(1, 2)$ satisfies both inequalities.
- Check the first: $2 > 2(1) - 1 = 1$? True. Check the second: $2 \leq -1 + 5 = 4$? True. Since $(1, 2)$ satisfies both inequalities, it lies in the solution region.
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Interpreting linear inequalities in context

- 10 A student has 50 dollars and wants to buy notebooks costing 6 dollars each, keeping at least 8 dollars in reserve. Write an inequality for the number of notebooks n the student can buy, and solve it.
- The amount spent, $6n$, must leave at least 8 dollars: $50 - 6n \geq 8$, so $-6n \geq -42$, giving $n \leq 7$ (dividing by a negative number flips the inequality). The student can buy at most 7 notebooks.
- 11 A factory produces at least 200 units per day but at most 350 units per day. Write a compound inequality for the daily production p , and state whether $p = 180$ is a valid production level.
- $200 \leq p \leq 350$. Since $180 < 200$, the value $p = 180$ does NOT satisfy the inequality — it is below the minimum required production level.
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Comparing rates of change between two linear functions

- 12 Function f is given by $f(x) = 3x + 2$. Function g is defined by the table: $x = 0, g = 1$ and $x = 2, g = 9$. Which function has the greater rate of change?
- f has a rate of change (slope) of 3. For g : slope $= \frac{9 - 1}{2 - 0} = \frac{8}{2} = 4$. Since $4 > 3$, function g has the greater rate of change.
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Finding the intersection of two lines graphically and algebraically

- 13 Two lines are given by $y = 2x - 1$ and $y = -x + 8$. Find their point of intersection, and explain what this point represents graphically.
- Set equal: $2x - 1 = -x + 8$, so $3x = 9$, giving $x = 3$. Then $y = 2(3) - 1 = 5$. The lines intersect at $(3, 5)$, which represents the single point where both lines cross on a graph — the unique solution to the system formed by the two equations.
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Absolute value equations and inequalities

- 14 Solve $|2x - 3| = 9$.
- Either $2x - 3 = 9$, giving $x = 6$, or $2x - 3 = -9$, giving $x = -3$. Solutions: $x = 6$ or $x = -3$.
- 15 Solve $|x + 4| < 6$, giving your answer as an interval.
- $-6 < x + 4 < 6$, so $-10 < x < 2$.

Writing a linear inequality from a verbal description

- 16** A shipping company charges 3 dollars per pound but offers free shipping if the package weighs less than 2 pounds. Write a piecewise description of the cost $C(w)$ as a function of weight w , and find the cost of a 5-pound package.

$C(w) = 0$ for $w < 2$ and $C(w) = 3w$ for $w \geq 2$. For a 5-pound package: $C(5) = 3(5) = 15$ dollars.

- 17** A parking garage charges 5 dollars for the first hour and 2 dollars for each additional hour. Write an equation for the total cost C after h hours (for $h \geq 1$), and find the cost for 6 hours.

$C(h) = 5 + 2(h - 1)$ for $h \geq 1$ (since only the hours after the first are charged at the additional rate). For 6 hours: $C(6) = 5 + 2(5) = 5 + 10 = 15$ dollars.

Graphing a system of inequalities and identifying the feasible region

- 18** A system requires $x + y \leq 10$ and $x \geq 2$ and $y \geq 0$. Determine which of the points $(3, 4)$ and $(8, 5)$ lie in the feasible (solution) region, checking each condition.

For $(3, 4)$: $3 + 4 = 7 \leq 10$ ✓, $3 \geq 2$ ✓, $4 \geq 0$ ✓ — all conditions satisfied, so $(3, 4)$ is in the feasible region. For $(8, 5)$: $8 + 5 = 13 \leq 10$? False — this point fails the first condition, so $(8, 5)$ is NOT in the feasible region.

Rate of change from a real-world linear model

- 19** A water tank starts with 500 gallons and drains at a constant rate. After 4 hours, it contains 340 gallons. Write a linear equation for the volume V after t hours, and predict when the tank will be empty.

Rate of change: $\frac{340 - 500}{4 - 0} = \frac{-160}{4} = -40$ gallons per hour. Equation: $V = 500 - 40t$. The tank is empty when $V = 0$: $500 - 40t = 0$, so $t = 12.5$ hours.